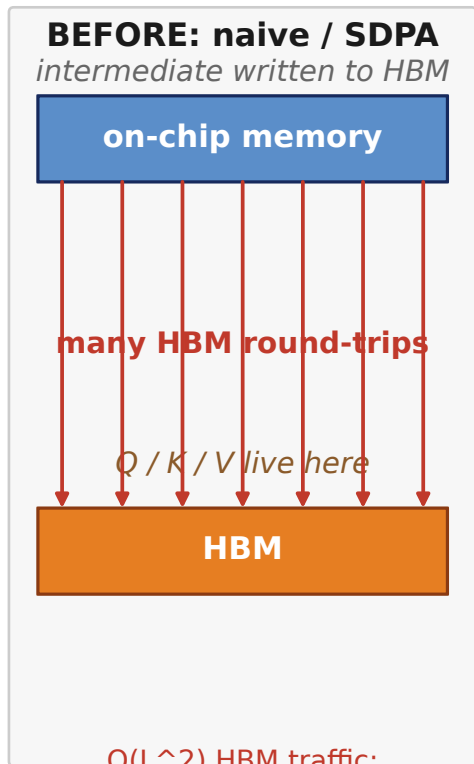
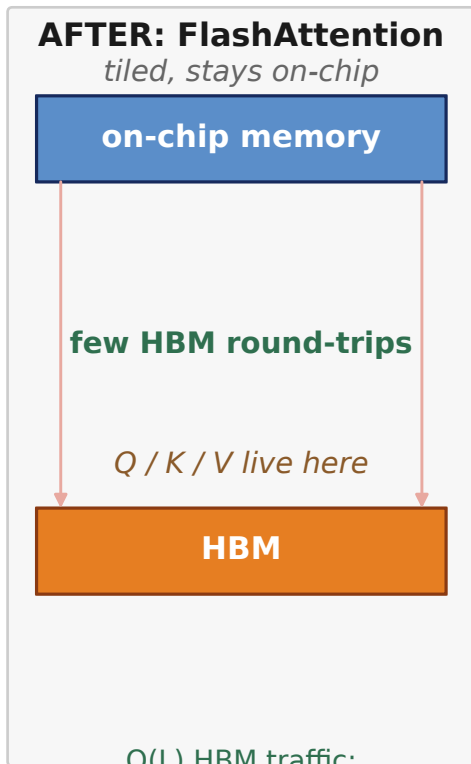


# FlashAttention: the same math, far less HBM traffic



$O(L^2)$  HBM traffic:  
every query re-reads  
the row-block



$O(L)$  HBM traffic:  
each tile read once,  
softmax online

**same  $Q \cdot K^T \cdot V$  math and same result —  
the gain is fewer HBM round-trips, not cheaper attention**